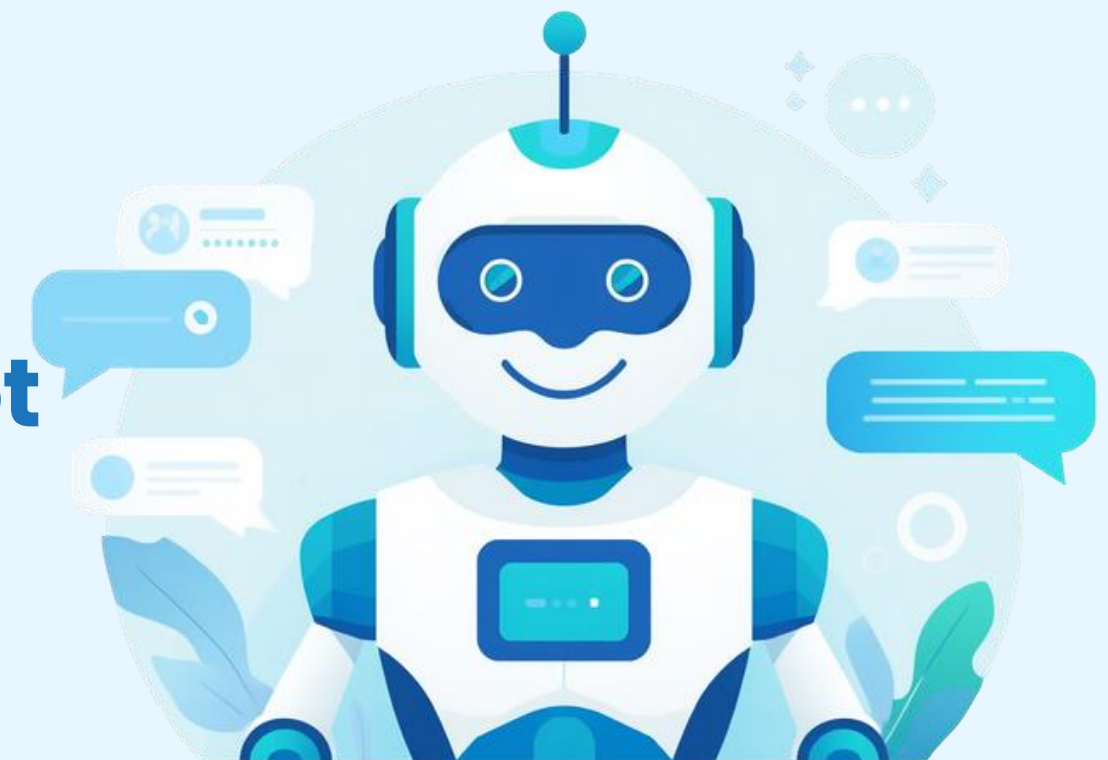


AI CHATBOT FOR STUDENT SUPPORT SERVICES

Enhancing Student Experience through Intelligent
Automation

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Date-14 July, 2026
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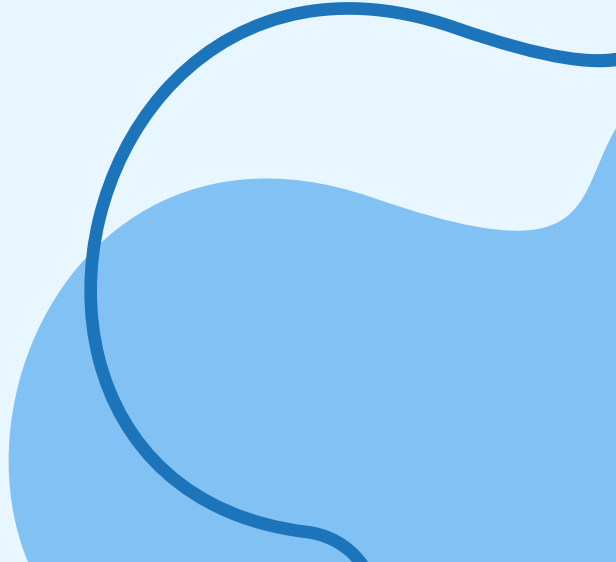
PROJECT OVERVIEW



Increasing volume of student queries across departments — admissions, fees, exams, scholarships, library, hostel, and placements — demands a smarter, faster support system.

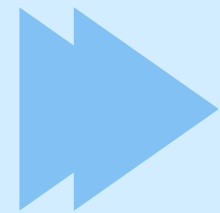


There is a critical need for instant, 24/7 support to improve efficiency and student satisfaction. We introduce an AI-based chatbot solution to address these growing challenges.

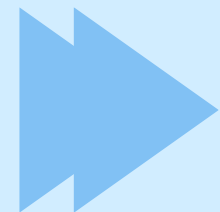


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SOLUTION SUMMARY



AI-powered chatbot providing instant, automated responses to student queries. Uses Machine Learning (Naive Bayes) and NLP (TF-IDF) for accurate query classification and response generation.



24/7 availability ensures students receive support anytime, reducing staff workload and significantly accelerating response times across all support departments.

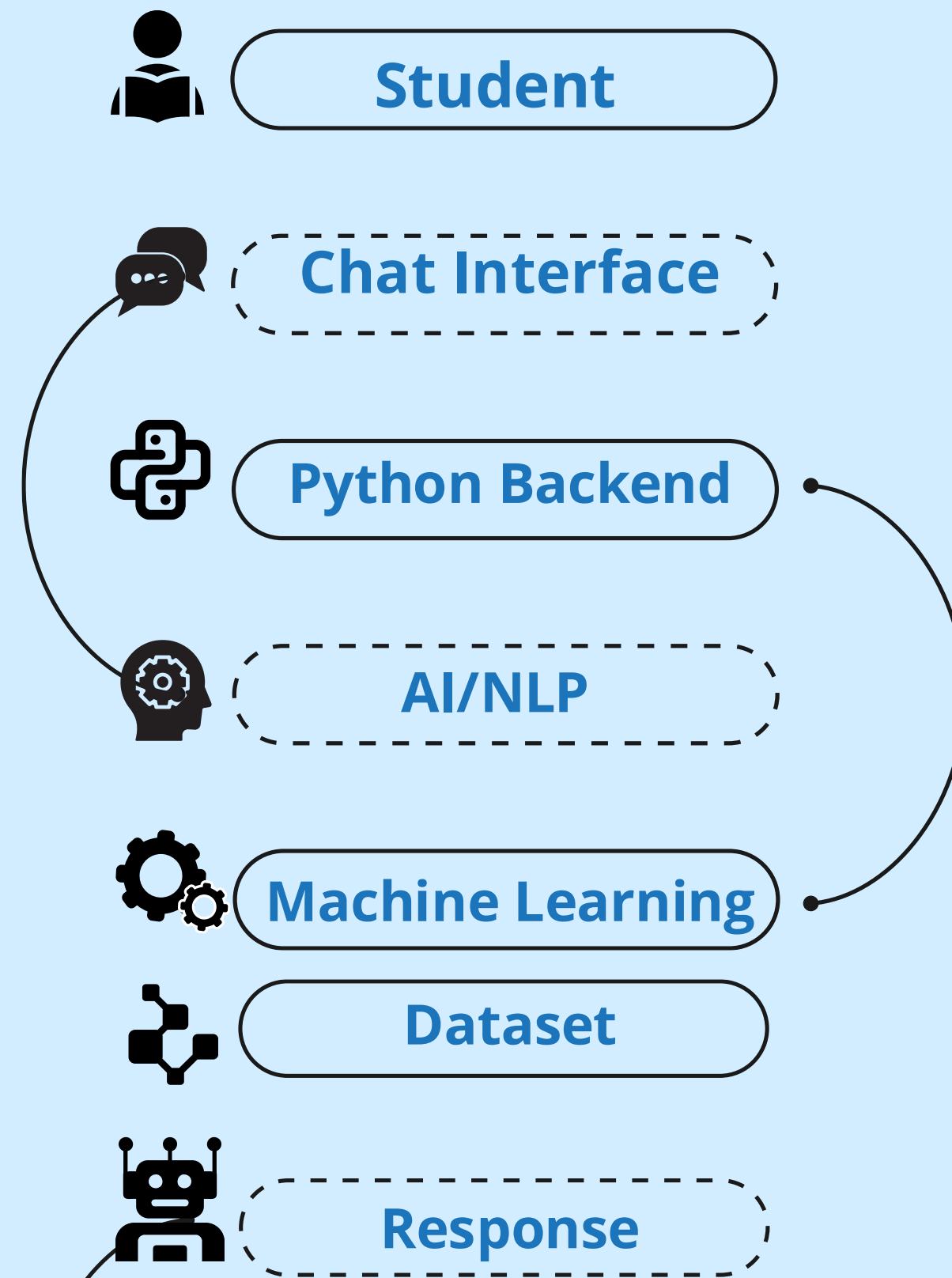
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SYSTEM ARCHITECTURE

The chatbot follows a modular, scalable pipeline. A student submits a query via the Gradio chat interface, which forwards it to the Python backend. NLP processing applies TF-IDF vectorization to extract key features. The Naive Bayes ML model then classifies the query intent. The matched intent is used to retrieve the appropriate response from the structured knowledge dataset. The final response is returned to the student instantly. Each component is independently maintainable, enabling easy updates and scalability.



SYSTEM ARCHITECTURE



TECHNOLOGIES USED



Python programming language as the core backend
Google Colab as the development and testing environment
Scikit-learn for the Naive Bayes ML classification model

Pandas for data loading, cleaning, and processing
Gradio for building a user-friendly chat interface
TF-IDF (NLP) for feature extraction and query understanding

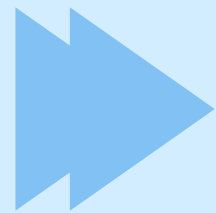
TECHNOLOGIES IN ACTION

This slide is reserved for a screenshot showcasing the technologies used in the AI Chatbot project.

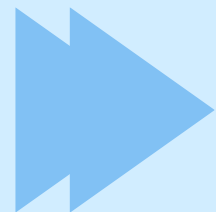
- Python development environment (Google Colab)
- Scikit-learn ML model implementation
- Pandas data processing pipeline
- Gradio chat interface in action



CHATBOT FEATURES

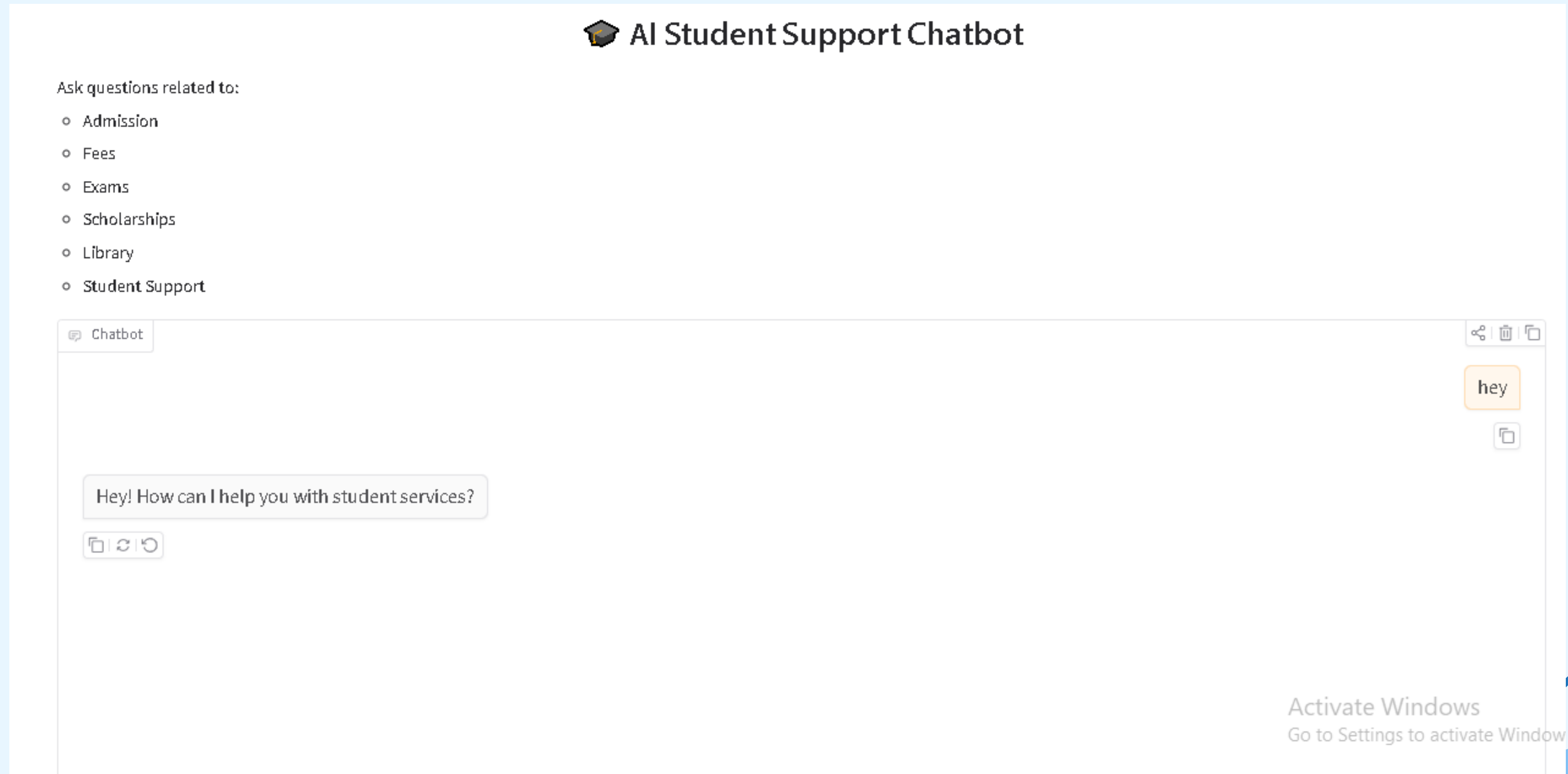


Instant responses to common student queries covering admissions, fees, exams, scholarships, library, hostel, and placements. User-friendly chat interface accessible anytime, anywhere.



Available 24/7 to provide round-the-clock support, reducing wait times and improving student satisfaction. Automated responses reduce workload on support staff significantly.

CHATBOT USER INTERFACE





SYSTEM WORKFLOW

Student submits a query via the chat interface. The Python backend receives the input and preprocesses the text by cleaning, tokenizing, and normalizing it for further analysis.

NLP applies TF-IDF vectorization to extract meaningful features from the preprocessed text. The Naive Bayes ML model then classifies the query intent with high accuracy.

The matched response is retrieved from the knowledge dataset based on the classified intent. The response is instantly sent back to the student through the chat interface.

BENEFITS OF THE AI CHATBOT

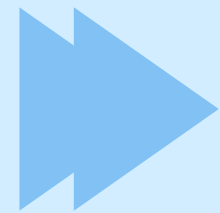


Faster response times compared to manual handling, reducing student wait times significantly. Reduces workload on support staff, freeing them to focus on complex issues requiring human attention.

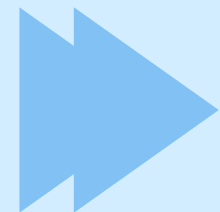


Available 24/7, improving accessibility for students at any time of day or night. Enhances overall student experience and satisfaction through instant, reliable support across all service areas.

LIMITATIONS



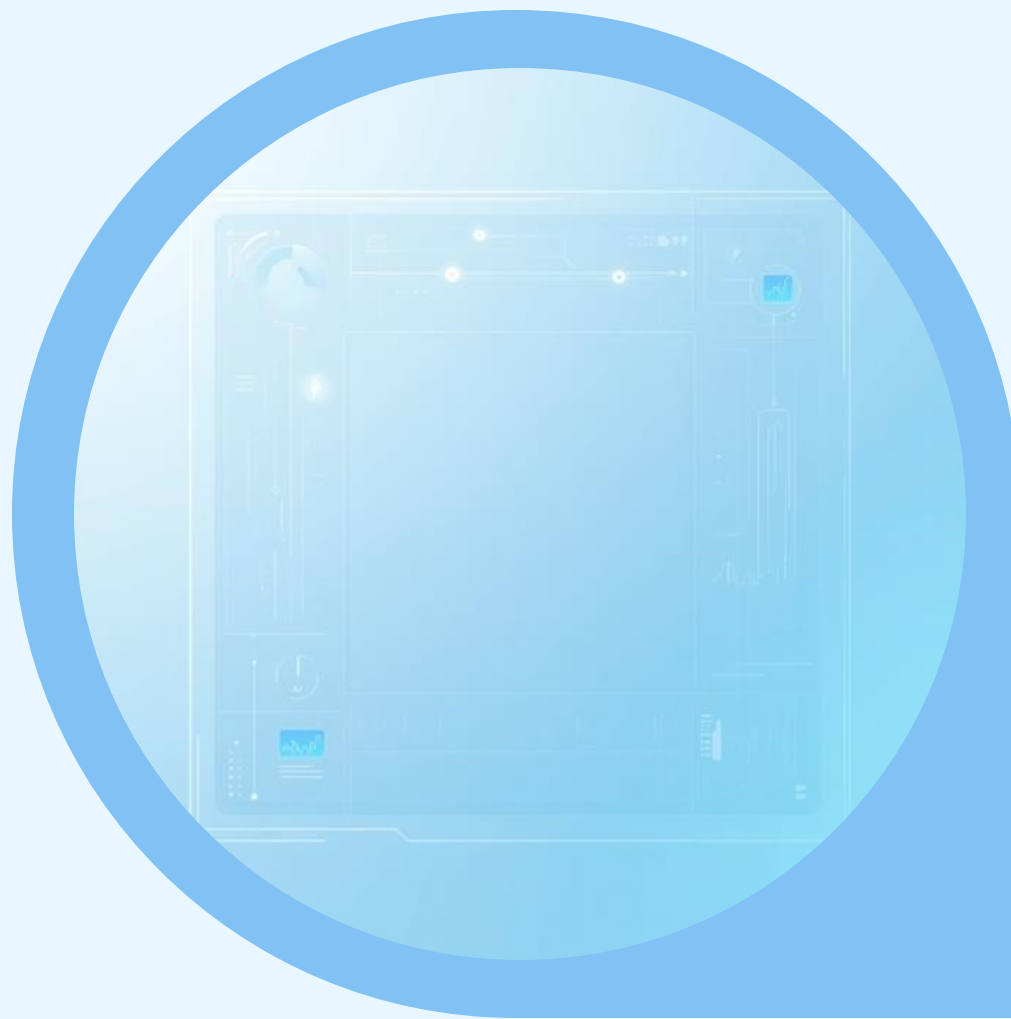
Accuracy depends heavily on the quality and scope of training data. Limited or biased datasets may result in incorrect query classification and poor response accuracy.



Cannot manage complex or personal issues requiring human intervention. Requires regular updates and ongoing maintenance to remain effective and relevant.

1

FUTURE ENHANCEMENTS



Voice-based interaction capabilities for hands-free access. Integration with student databases to deliver personalized responses tailored to individual student profiles and needs.

Development of a dedicated mobile app for wider accessibility. Support for multiple languages to serve diverse student populations across regions and backgrounds.

SUMMARY & CONCLUSION

”

The AI Chatbot is a scalable, intelligent solution for student support services. By integrating Machine Learning (Naive Bayes) and NLP (TF-IDF), it delivers fast, accurate responses 24/7. The system reduces staff workload, enhances accessibility, and improves the overall student experience — with strong potential for future growth and expansion.

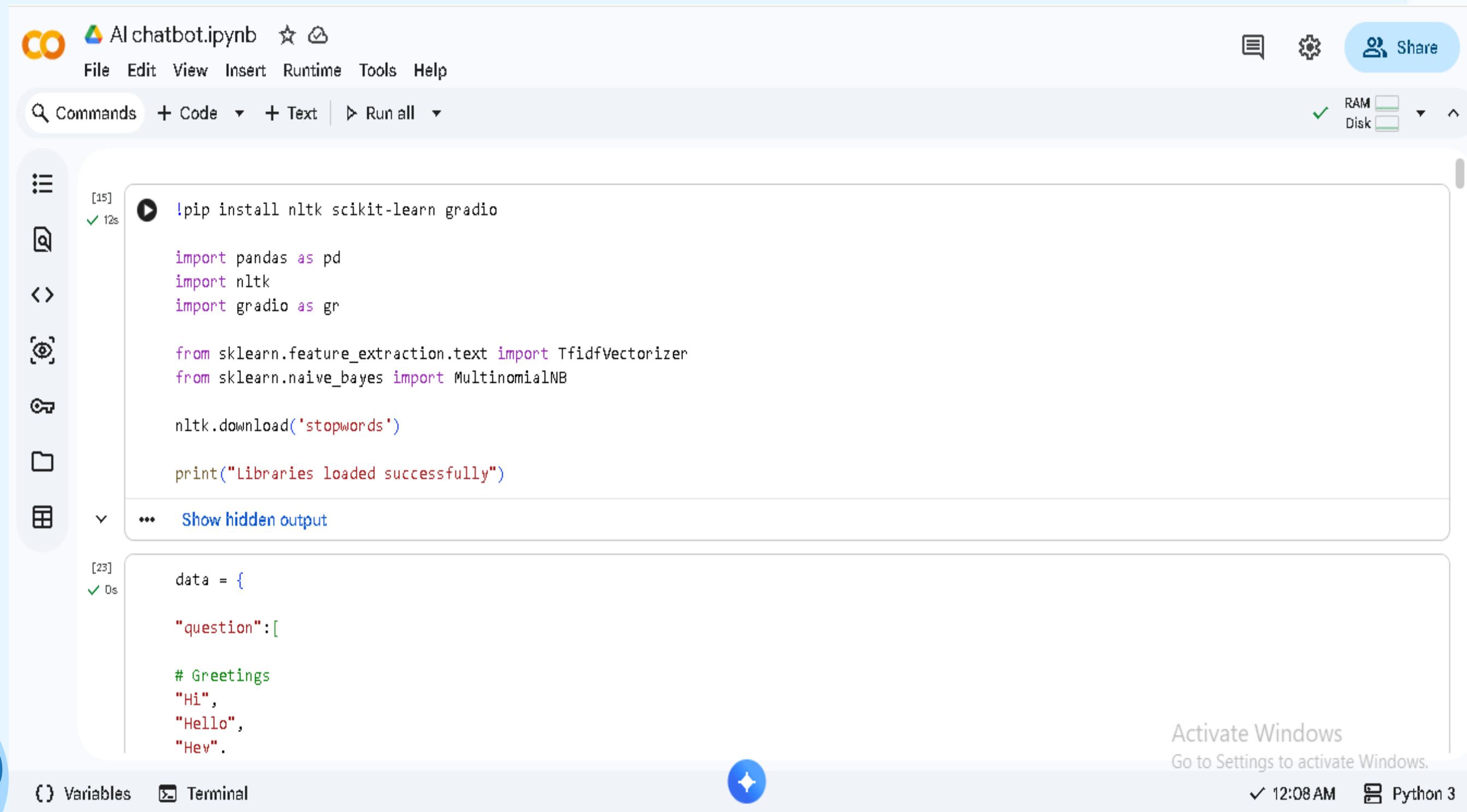




NEXT STEPS

1. Expand the knowledge dataset to cover a wider range of student queries across all departments, improving chatbot accuracy and response coverage.
2. Collect user feedback systematically and apply iterative improvements to the ML model and response database to enhance performance over time.
3. Conduct a pilot deployment in a real student environment, monitor chatbot performance, and address issues before full-scale rollout.

DEMO SCREENSHOT 1



The screenshot displays a Jupyter Notebook titled "AI chatbot.ipynb". The interface includes a top menu bar with options like File, Edit, View, Insert, Runtime, Tools, and Help. Below the menu is a toolbar with icons for running code, adding code cells, and a "Run all" button. The notebook contains two code cells. The first cell, labeled [15], shows the installation of NLTK and Scikit-Learn, followed by imports for pandas, nltk, and gradio, and the download of NLTK stopwords. The second cell, labeled [23], defines a dataset with a "question" list containing greetings like "Hi", "Hello", and "Hev". The bottom status bar shows the current time as 12:08 AM and the Python version as Python 3.

```
[15] !pip install nltk scikit-learn gradio

import pandas as pd
import nltk
import gradio as gr

from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB

nltk.download('stopwords')

print("Libraries loaded successfully")

... Show hidden output

[23] data = {

    "question":[

        # Greetings
        "Hi",
        "Hello",
        "Hev".
```

Activate Windows
Go to Settings to activate Windows.

✓ 12:08 AM Python 3



DEMO SCREENSHOT 2

```
Hey ,
"Hi!",
"Good morning",
"Good afternoon",
"Good evening",
"Are you there?",
"Can I talk to you?",
"Help me",
"I need help",
"Can you help me?",

# Admission
"How can I apply for admission?",
"What is the admission process?",
"When will admission start?",
"How to get admission details?",
"Where can I apply for admission?",
"How to take admission?",
"Admission form details",
"What documents are required for admission?",
"When does admission open?",
"I want admission information",

# Fees
"What is the fee structure?",
```

```
}

df = pd.DataFrame(data)

df.head()
```

	question	intent	answer
0	Hi	greeting	Hey! How can I help you with student services?
1	Hello	greeting	Hello! How can I assist you today?
2	Hey	greeting	Hi! Feel free to ask me any student-related qu...
3	Hi	greeting	Hi there! How can I help you?
4	Good morning	greeting	Good morning! How may I help you?

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
x = df["question"]
y = df["intent"]
```

DEMO SCREENSHOT 3

```
def chat_function(message, history):  
  
    response = chatbot_response(message)  
  
    return response  
  
demo = gr.ChatInterface(  
  
    fn=chat_function,  
  
    title="🎓 AI Student Support Chatbot",  
  
    description="""  
Ask questions related to:  
- Admission  
- Fees  
- Exams  
- Scholarships  
- Library  
- Student Support  
""")
```



DEMO SCREENSHOT 4

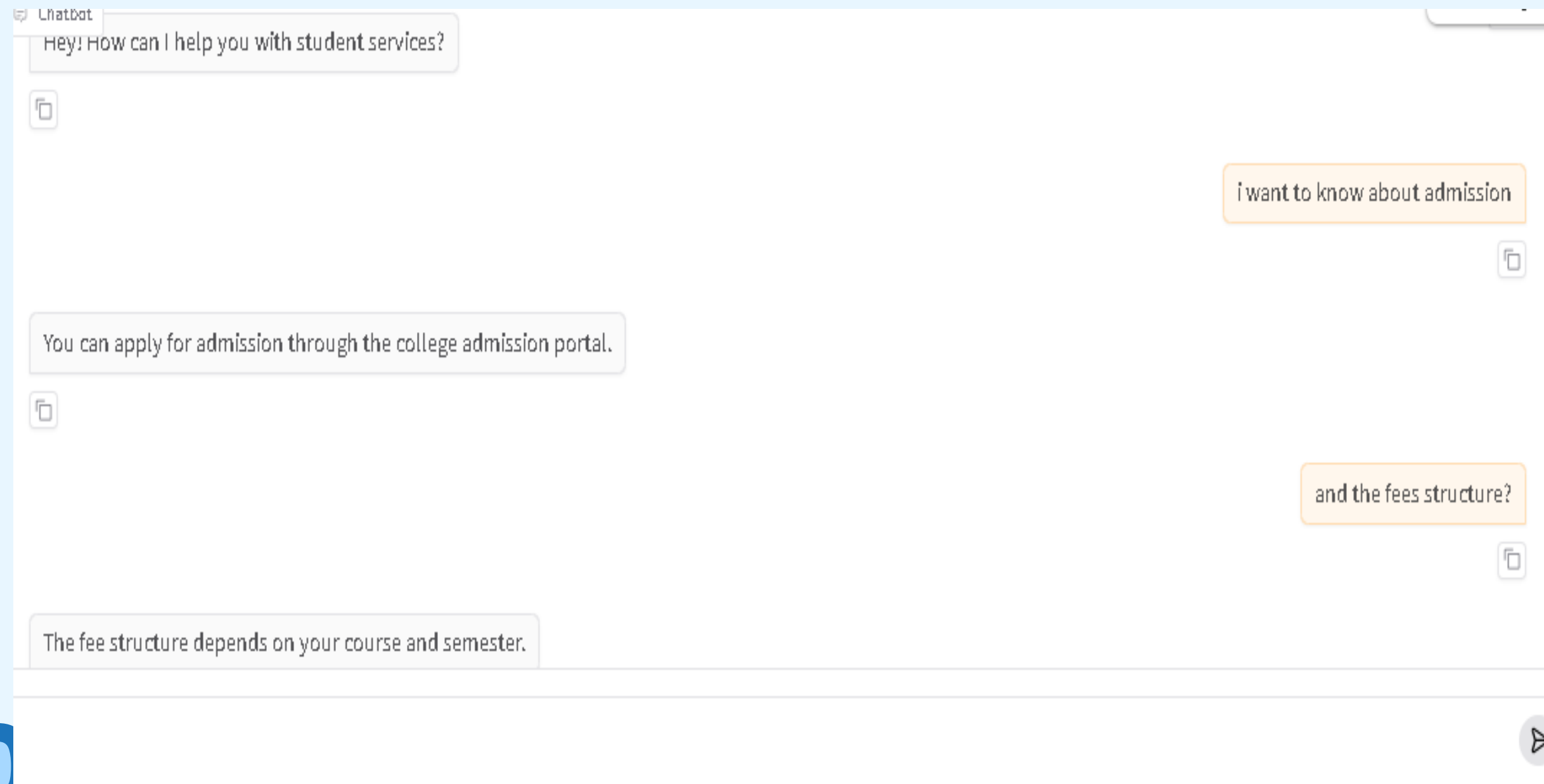
• It looks like you are running Gradio on a hosted Jupyter notebook, which requires `share=True`. Automatically setting `share=True` (you can turn t

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()

* Running on public URL: <https://d0359ffb7e08f3becc.gradio.live>

This share link is temporary and will last for up to 1 week (best effort). For free permanent hosting and GPU upgrades, run `gradio deploy` from t

DEMO SCREENSHOT 5



THANK YOU!

